

* Computer Network Lab - 02

Q.1

What are the different protocols you observe at the following layers of the protocol stack?

- A. Application layer
- B. Transport layer
- C. Network layer.

Ans:

- Application layer.

→ The Application layer protocols observed in the packet capture are :

- HTTP (Hypertext Markup Language)
- PNG (Portable Network Graphics)

- Transport layer

→ The Transport layer protocols observed is :

- TCP (Transmission Control Protocol)

TCP is responsible for reliable communication b/w the client and server

• Network layer

→ The network layer protocols observed are:

- IPv4 (Internet protocol version 4)
- IPv6 (Internet protocol version 6)

These protocols are responsible for logical addressing and routing of packets.

Q.2

What is the total amount of data being received for the following cases?

(A) when you access <http://neverssl.com>

Ans: From the HTTP response packet.
content-length = 1900 bytes

The total file data received is: 3961 bytes

As: 3961 Bytes.

(B) what is the total amount of data received when access <http://www.cornell.edu>.

→ no ans capture

Q.3

Answer the following when you access the site `http://neverssl.com/`

(A) How many HTTP GET requests do you observe? List down the GET request.

A total 18 HTTP GET requests were observed in the packet capture.

The main GET request are

1. GET /connecttest.txt HTTP/2.0

2. GET / HTTP/2.0

3. GET /online HTTP/2.0

4. GET /online / HTTP/2.0

5. GET /favicon HTTP/2.0

several requests for /connecttest.txt were repeated during this capture.

(B) What version of HTTP the server is running?

Ans:

The server is using HTTP/2.0

The response packet shows: HTTP/2.0 200

The web server identified in the response is Apache/2.4.66

(c) For each of the HTTP GET request, find out:

(i) The total number of TCP segments being received.

the HTTP response was reassembled from 2 TCP segments.

This can be seen from:

Reassembled TCP segments (2273 bytes)
segment count = 2.

Therefore, the total number of TCP segments received was 2 TCP segments.

(ii) Find the total amount of data being received in the corresponding HTTP response message.

Ans: The HTTP response contains:
File Data : 3961 bytes

Therefore, the total amount of data received ^{is} 3961 bytes.

(d) Identify TCP three way handshake used to establish the connection with <http://neverssl.com>

Ans: The TCP connection was established using the standard three way.

SYN

Packet No. 3213, seq = 0

SYN-ACK

Packet No. 3750, seq = 0, Ack = 1

ACK

Packet No. 3751, seq = 1, Ack = 1